

Analysis of Large Graphs: Link Analysis, PageRank

Mining of Massive Datasets

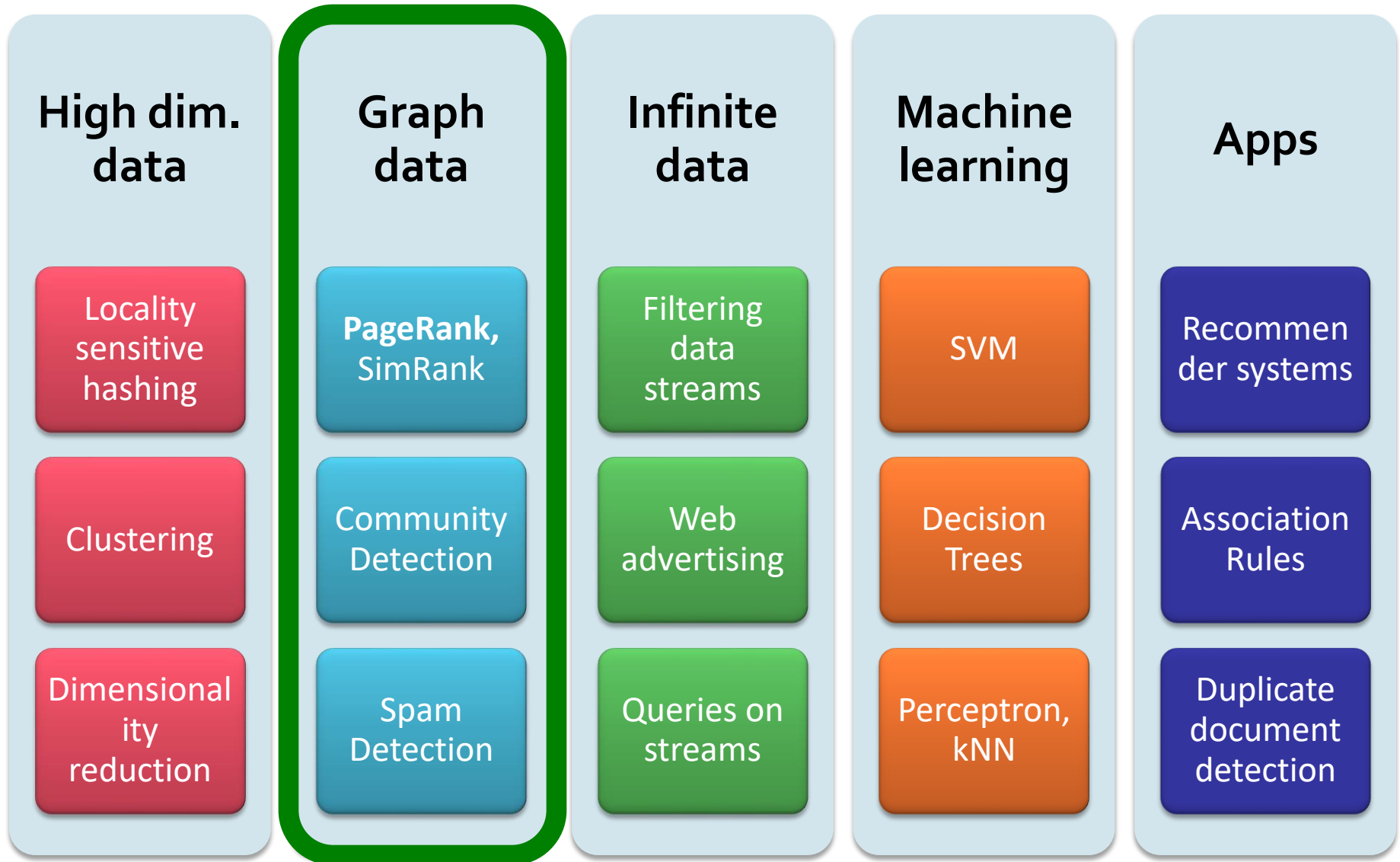
Jure Leskovec, Anand Rajaraman, Jeff Ullman

Stanford University

<http://www.mmds.org>



New Topic: Graph Data!



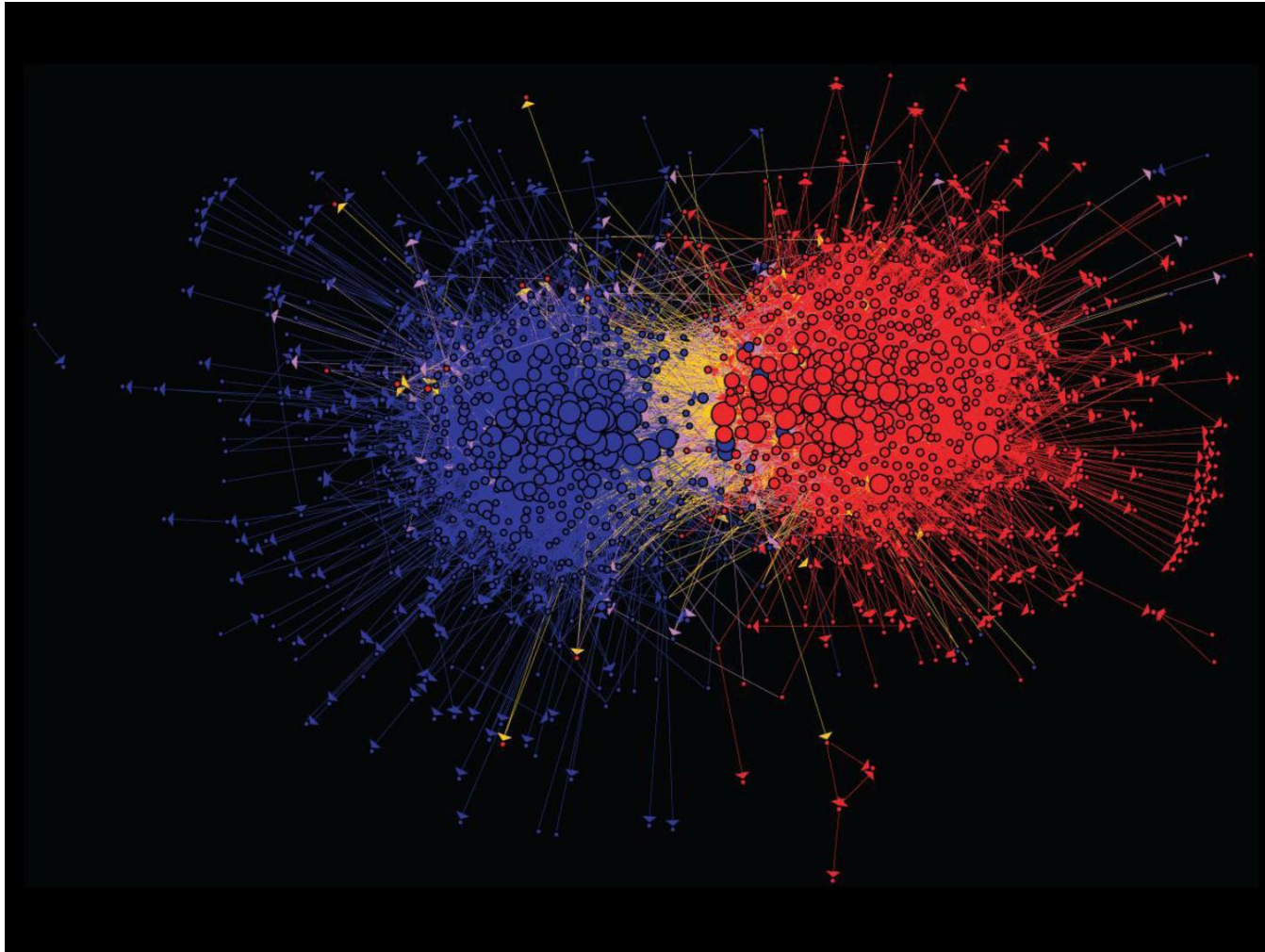
Graph Data: Social Networks



Facebook social graph

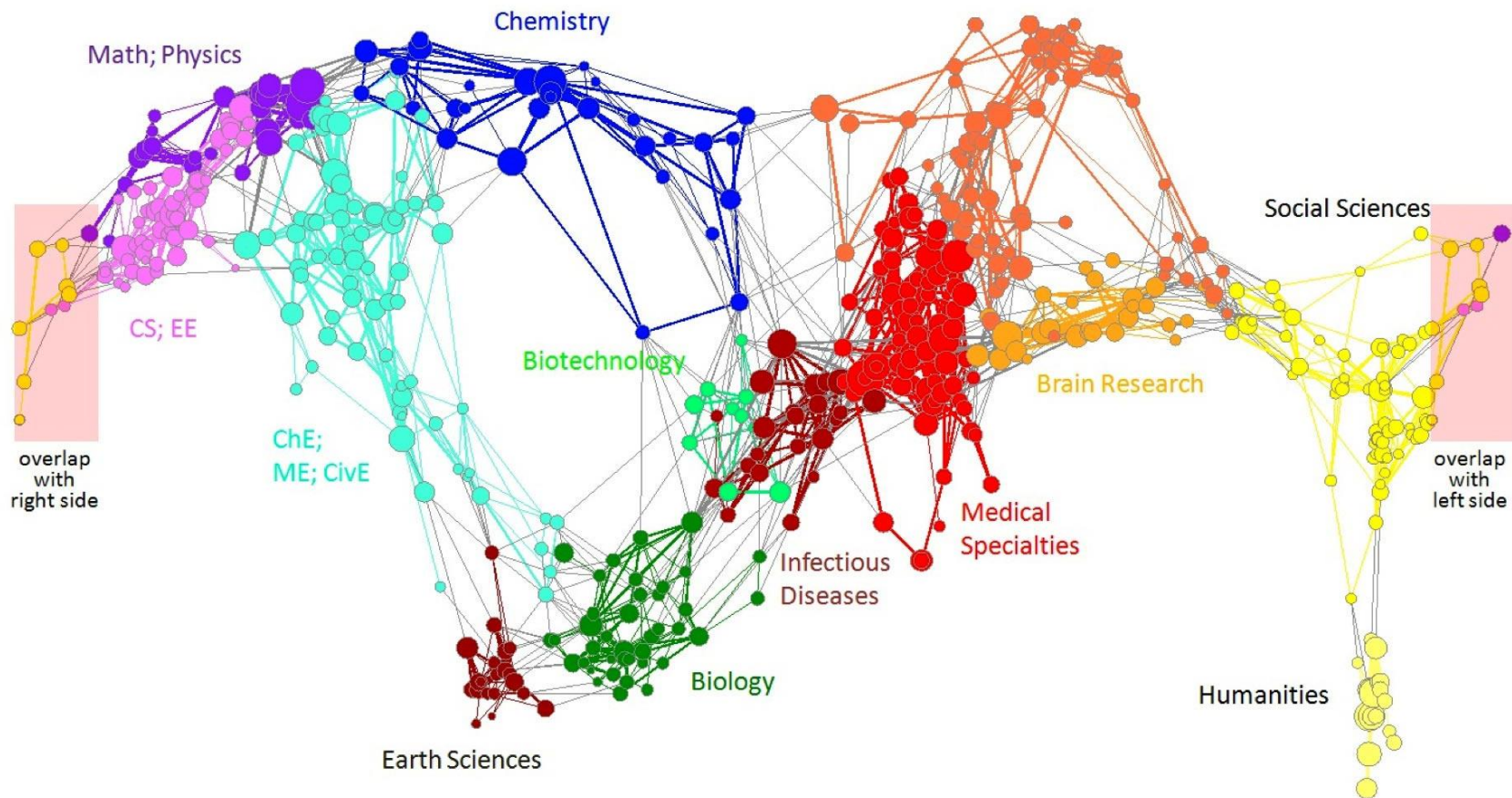
4-degrees of separation [Backstrom-Boldi-Rosa-Ugander-Vigna, 2011]

Graph Data: Media Networks



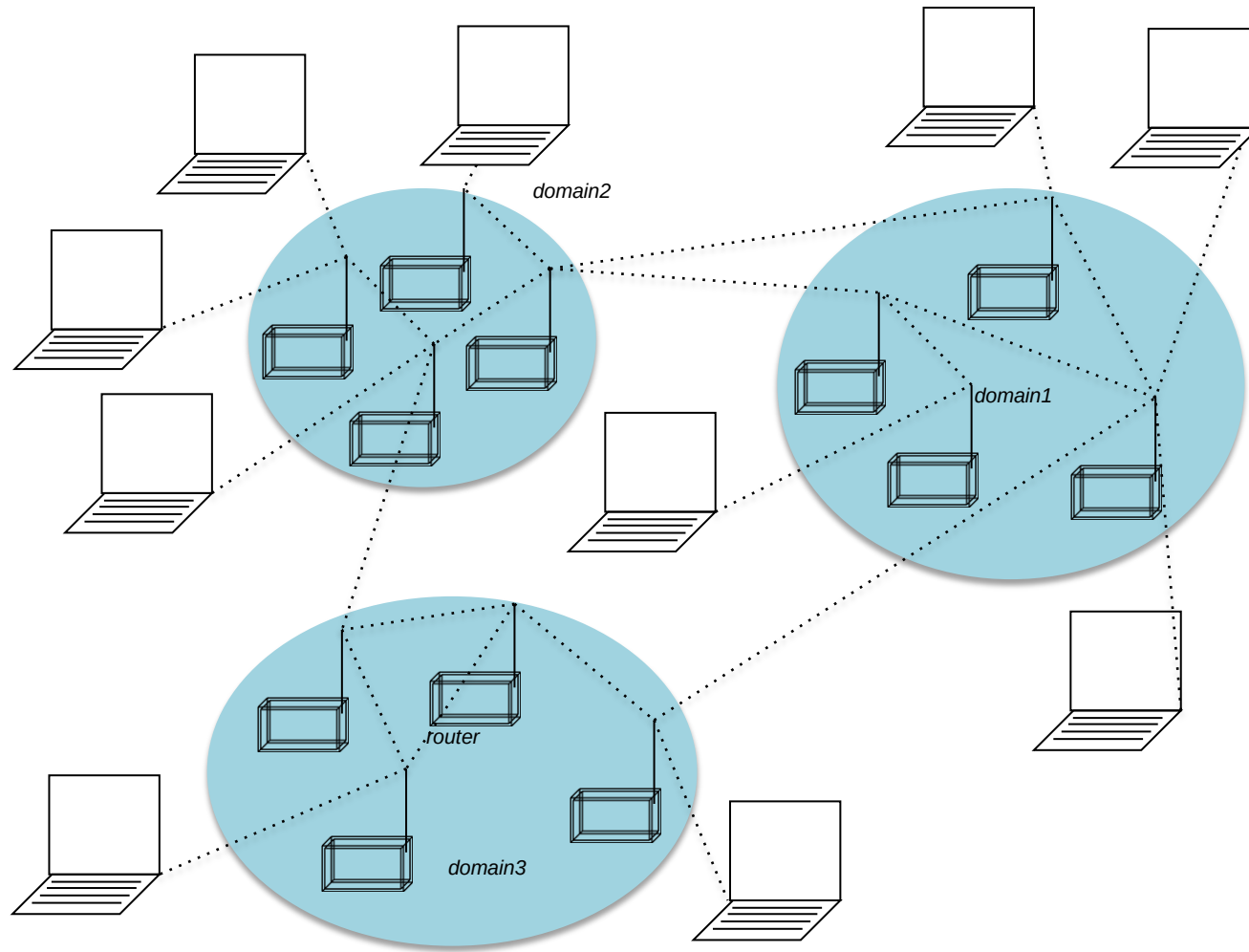
Connections between political blogs
Polarization of the network [Adamic-Glance, 2005]

Graph Data: Information Nets



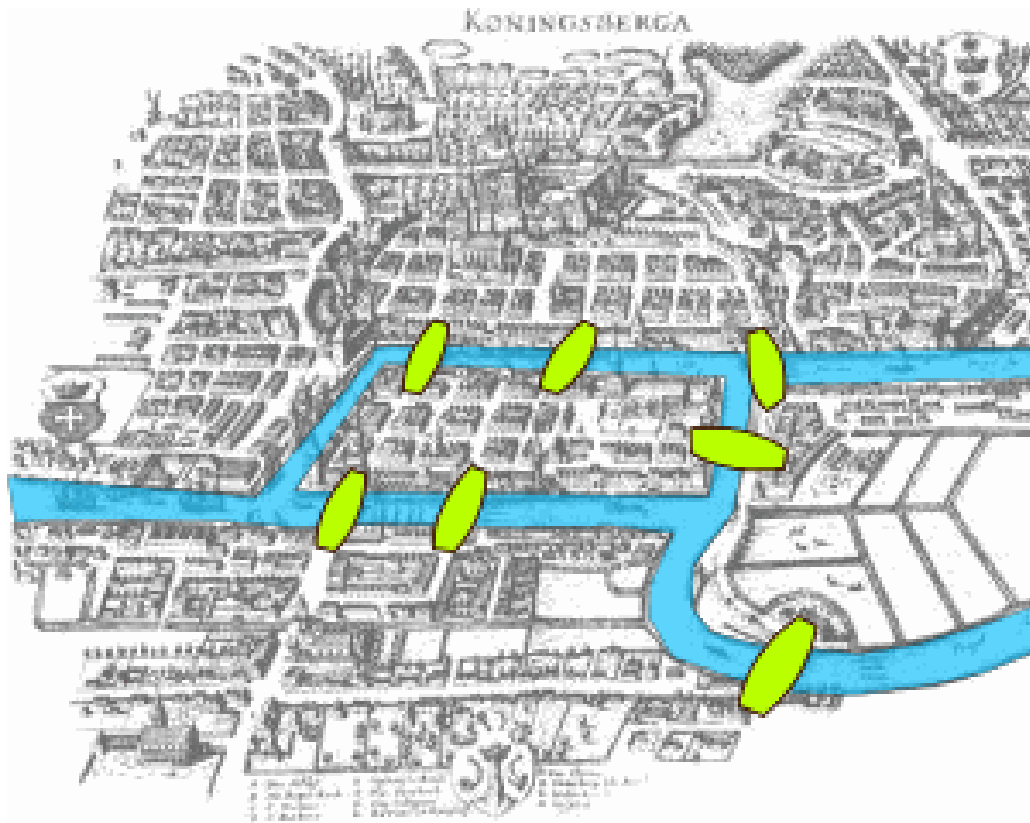
Citation networks and Maps of science
[Börner et al., 2012]

Graph Data: Communication Nets



Internet

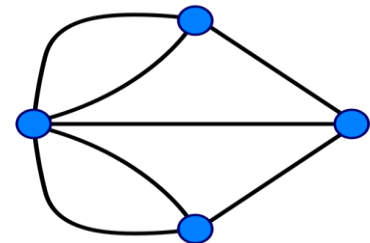
Graph Data: Technological Networks



Seven Bridges of Königsberg

[Euler, 1735]

Return to the starting point by traveling each link of the graph once and only once.



Web as a Graph

- **Web as a directed graph:**
 - **Nodes: Webpages**
 - **Edges: Hyperlinks**

I teach a
class on
Networks.

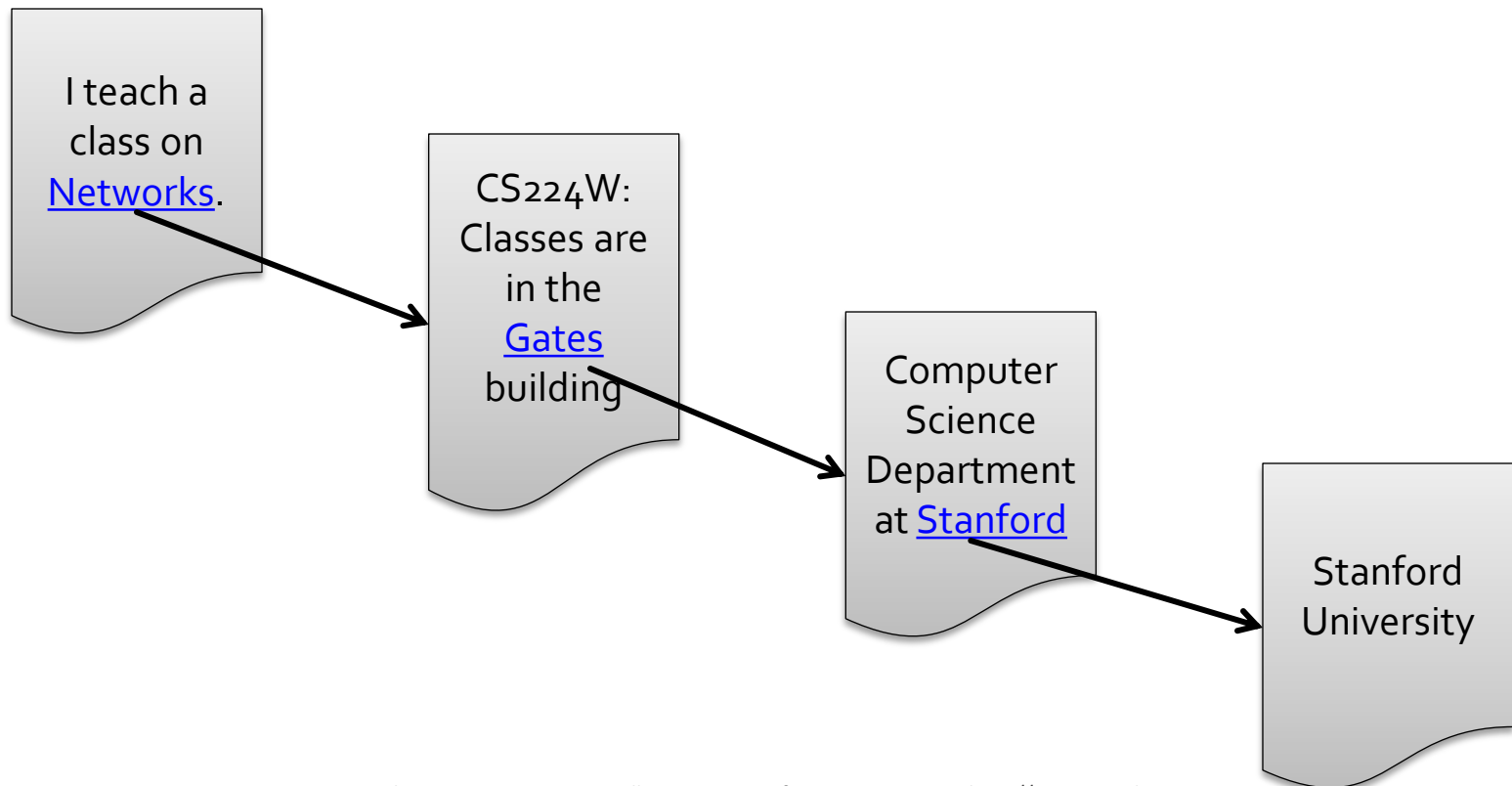
CS224W:
Classes are
in the
Gates
building

Computer
Science
Department
at Stanford

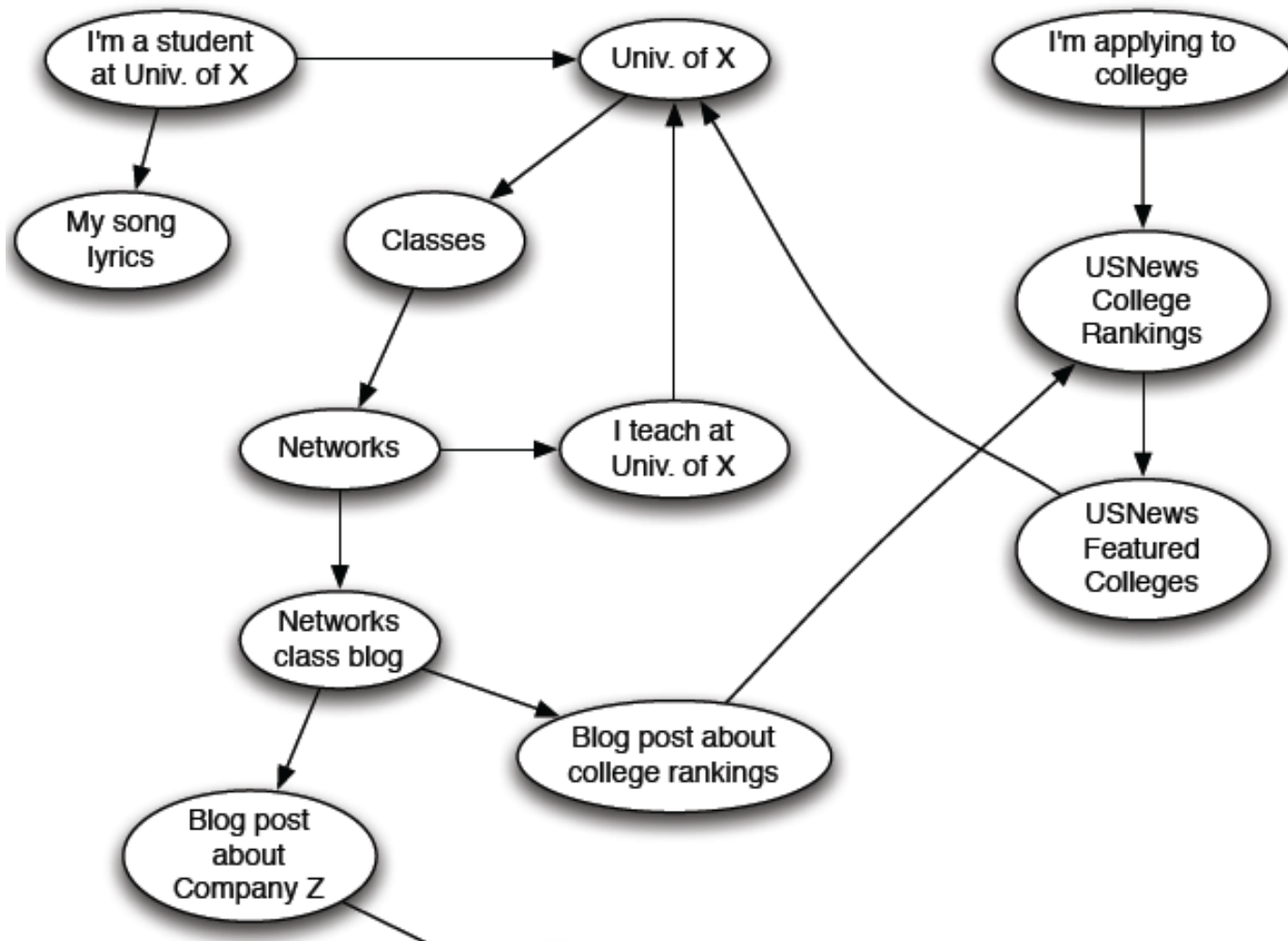
Stanford
University

Web as a Graph

- **Web as a directed graph:**
 - **Nodes: Webpages**
 - **Edges: Hyperlinks**



Web as a Directed Graph



Broad Question

- **How to organize the Web?**
- **First try: Human curated Web directories**
 - Yahoo, DMOZ, LookSmart
- **Second try: Web Search**
 - **Information Retrieval** investigates:
Find relevant docs in a small and trusted set
 - Newspaper articles, Patents, etc.
 - **But:** Web is **huge**, full of untrusted documents, random things, web spam, etc.



Web Search: 2 Challenges

2 challenges of web search:

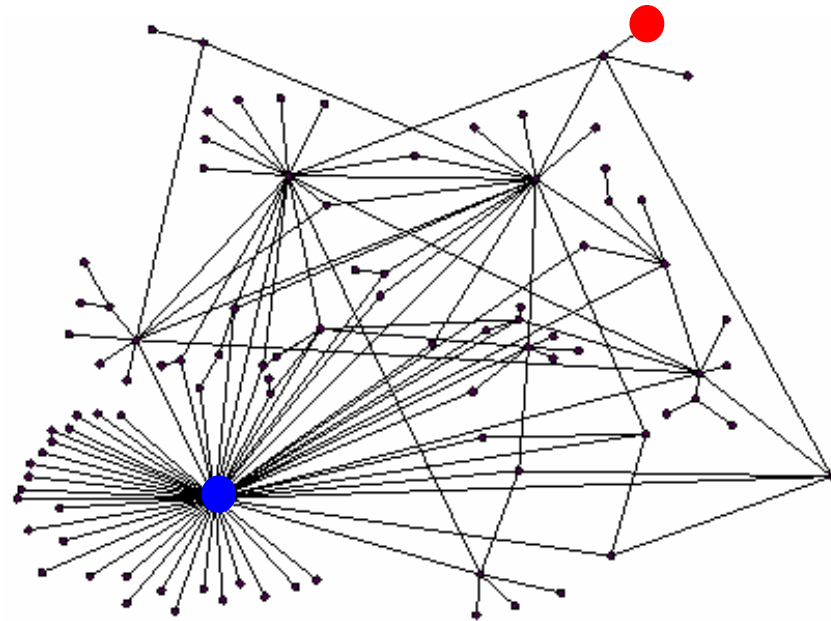
- (1) Web contains many sources of information
Who to “trust”?
 - **Trick:** Trustworthy pages may point to each other!
- (2) What is the “best” answer to query “newspaper”?
 - No single right answer
 - **Trick:** Pages that actually know about newspapers might all be pointing to many newspapers

Ranking Nodes on the Graph

- All web pages are not equally “important”

www.joe-schmoe.com vs. www.stanford.edu

- There is large diversity in the web-graph node connectivity.
Let's rank the pages by the link structure!



Link Analysis Algorithms

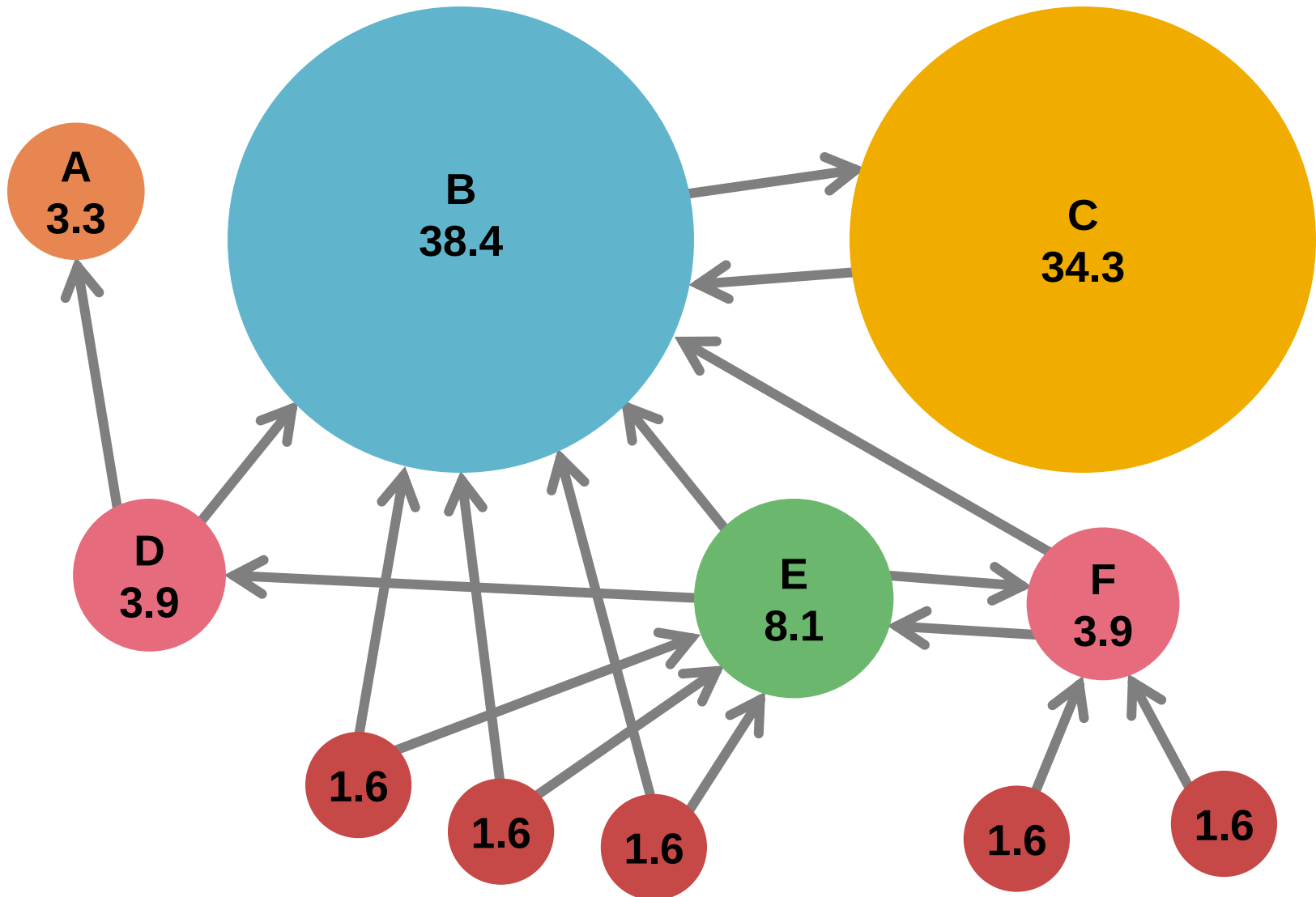
- We will cover the following **Link Analysis approaches** for computing **importances** of nodes in a graph:
 - Page Rank
 - Topic-Specific (Personalized) Page Rank
 - Web Spam Detection Algorithms

PageRank: The “Flow” Formulation

Links as Votes

- **Idea: Links as votes**
 - Page is more important if it has more links
 - In-coming links? Out-going links?
- **Think of in-links as votes:**
 - www.stanford.edu has 23,400 in-links
 - www.joe-schmoe.com has 1 in-link
- **Are all in-links are equal?**
 - Links from important pages count more
 - Recursive question!

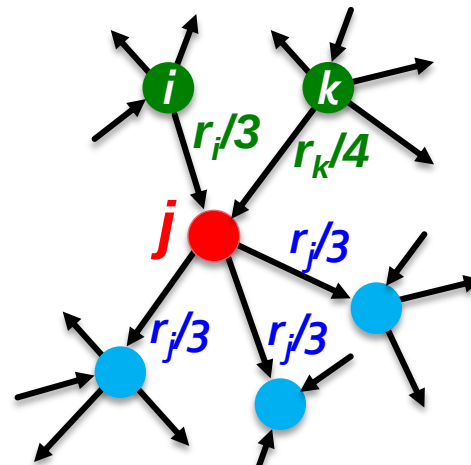
Example: PageRank Scores



Simple Recursive Formulation

- Each link's vote is proportional to the **importance** of its source page
- If page j with importance r_j has n out-links, each link gets r_j/n votes
- Page j 's own importance is the sum of the votes on its in-links

$$r_j = r_i/3 + r_k/4$$



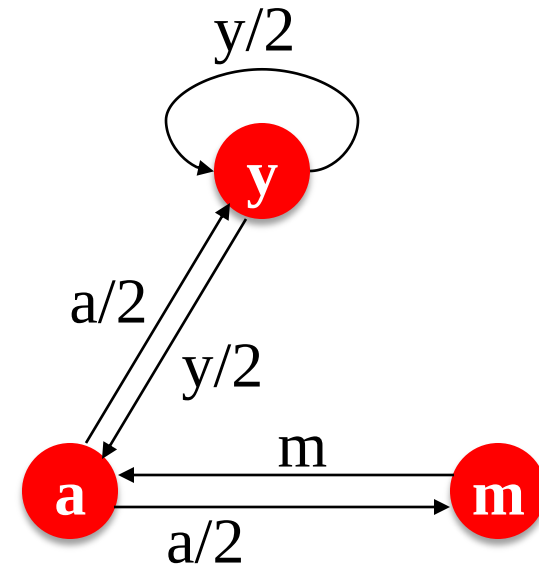
PageRank: The “Flow” Model

- A “vote” from an important page is worth more
- A page is important if it is pointed to by other important pages
- Define a “rank” r_j for page j

$$r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$$

d_i ... out-degree of node i

The web in 1839



“Flow” equations:

$$r_y = r_y/2 + r_a/2$$

$$r_a = r_y/2 + r_m$$

$$r_m = r_a/2$$

Solving the Flow Equations

- 3 equations, 3 unknowns,
no constants

- No unique solution
- All solutions equivalent modulo the scale factor

- **Additional constraint forces uniqueness:**

- $r_y + r_a + r_m = 1$

- **Solution:** $r_y = \frac{2}{5}, r_a = \frac{2}{5}, r_m = \frac{1}{5}$

- Gaussian elimination method works for small examples, but we need a better method for large web-size graphs
- **We need a new formulation!**

Flow equations:

$$r_y = r_y/2 + r_a/2$$

$$r_a = r_y/2 + r_m$$

$$r_m = r_a/2$$

PageRank: Matrix Formulation

- **Stochastic adjacency matrix M**

- Let page i has d_i out-links
- If $i \rightarrow j$, then $M_{ji} = \frac{1}{d_i}$ else $M_{ji} = 0$
 - M is a **column stochastic matrix**
 - Columns sum to 1

- **Rank vector r** : vector with an entry per page

- r_i is the importance score of page i
- $\sum_i r_i = 1$

- **The flow equations can be written**

$$r = M \cdot r$$

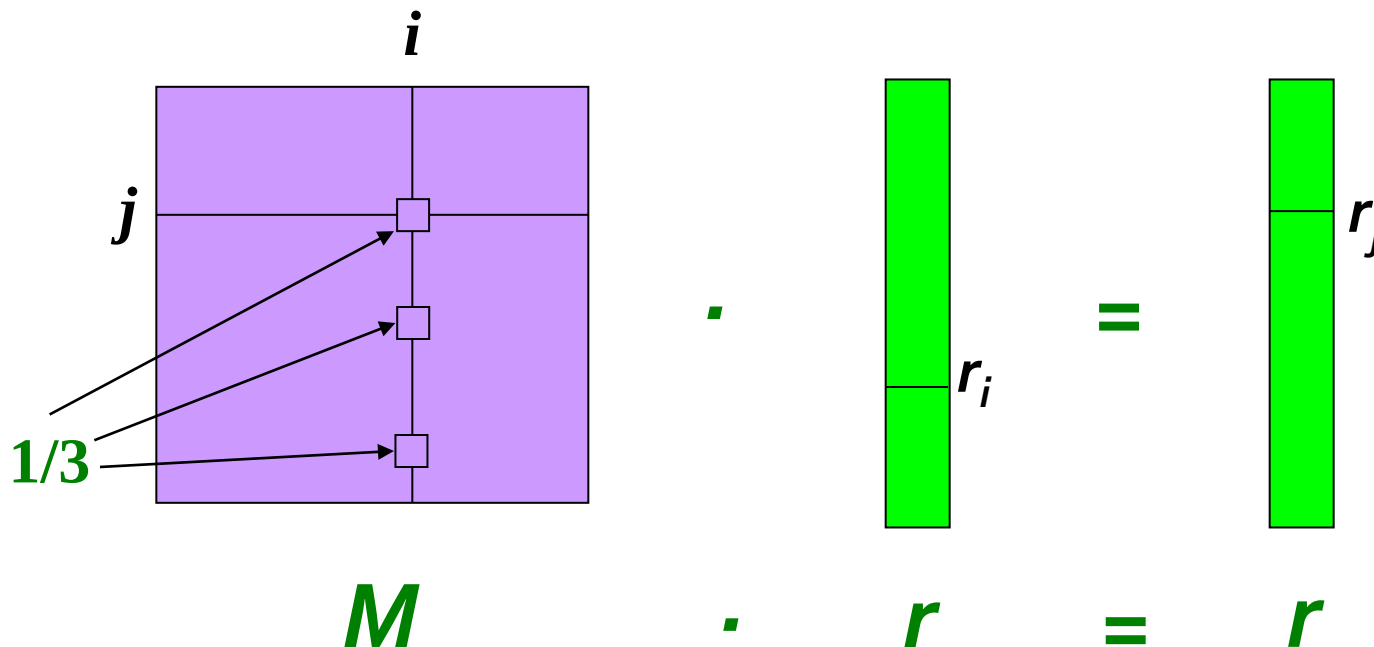
$$r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$$

Example

- Remember the flow equation: $r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
- Flow equation in the matrix form

$$M \cdot r = r$$

- Suppose page i links to 3 pages, including j



Eigenvector Formulation

- The flow equations can be written

$$\mathbf{r} = \mathbf{M} \cdot \mathbf{r}$$

- So the **rank vector** $\mathbf{r}$ is an **eigenvector** of the stochastic web matrix $\mathbf{M}$

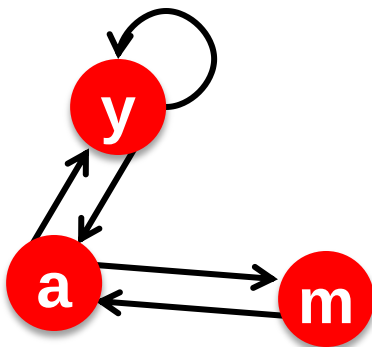
- In fact, its first or principal eigenvector, with corresponding eigenvalue **1**
 - Largest eigenvalue of $\mathbf{M}$ is **1** since $\mathbf{M}$ is column stochastic (with non-negative entries)
 - We know $\mathbf{r}$ is unit length and each column of $\mathbf{M}$ sums to one, so $\mathbf{M}\mathbf{r} \leq \mathbf{1}$

NOTE: $\mathbf{x}$ is an eigenvector with the corresponding eigenvalue λ if:

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

- **We can now efficiently solve for $\mathbf{r}$!**
The method is called Power iteration

Example: Flow Equations & M



	y	a	m
y	1/2	1/2	0
a	1/2	0	1
m	0	1/2	0

$$r = M \cdot r$$

$$r_y = r_y/2 + r_a/2$$

$$r_a = r_y/2 + r_m$$

$$r_m = r_a/2$$

$$\begin{bmatrix} y \\ a \\ m \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} y \\ a \\ m \end{bmatrix}$$

Power Iteration Method

- Given a web graph with n nodes, where the nodes are pages and edges are hyperlinks
- **Power iteration:** a simple iterative scheme

- Suppose there are N web pages

- Initialize: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$

- Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M} \cdot \mathbf{r}^{(t)}$

- Stop when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$

$\|\mathbf{x}\|_1 = \sum_{1 \leq i \leq N} |x_i|$ is the **L₁** norm

Can use any other vector norm, e.g., Euclidean

$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i}$$

d_i out-degree of node i

PageRank: How to solve?

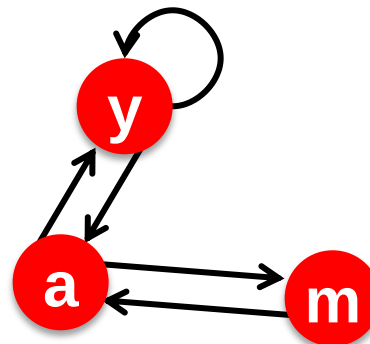
■ Power Iteration:

- Set $r_j = 1/N$
- 1: $r'_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
- 2: $r = r'$
- Goto 1

■ Example:

$$\begin{bmatrix} r_y \\ r_a \\ r_m \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

Iteration 0, 1, 2, ...



	y	a	m
y	1/2	1/2	0
a	1/2	0	1
m	0	1/2	0

$$r_y = r_y/2 + r_a/2$$

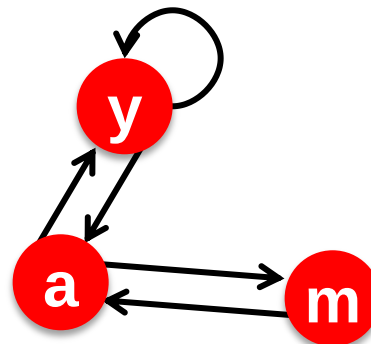
$$r_a = r_y/2 + r_m$$

$$r_m = r_a/2$$

PageRank: How to solve?

■ Power Iteration:

- Set $r_j = 1/N$
- 1: $r'_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
- 2: $r = r'$
- Goto 1



	y	a	m
y	1/2	1/2	0
a	1/2	0	1
m	0	1/2	0

$$\mathbf{r}_y = \mathbf{r}_y/2 + \mathbf{r}_a/2$$

$$\mathbf{r}_a = \mathbf{r}_y/2 + \mathbf{r}_m$$

$$\mathbf{r}_m = \mathbf{r}_a/2$$

■ Example:

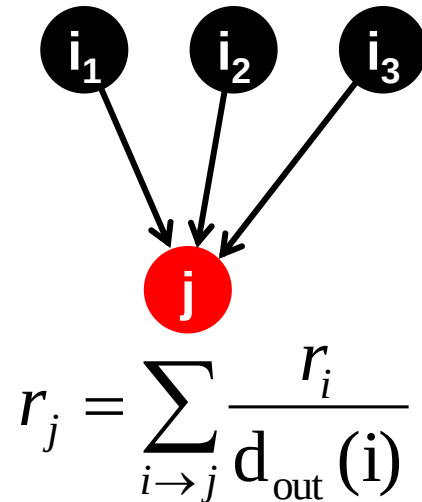
$$\begin{pmatrix} r_y \\ r_a \\ r_m \end{pmatrix} = \begin{pmatrix} 1/3 & 1/3 & 5/12 & 9/24 & & 6/15 \\ 1/3 & 3/6 & 1/3 & 11/24 & \dots & 6/15 \\ 1/3 & 1/6 & 3/12 & 1/6 & & 3/15 \end{pmatrix}$$

Iteration 0, 1, 2, ...

Random Walk Interpretation

- **Imagine a random web surfer:**

- At any time t , surfer is on some page i
- At time $t + 1$, the surfer follows an out-link from i uniformly at random
- Ends up on some page j linked from i
- Process repeats indefinitely



- **Let:**

- $\mathbf{p}(t)$... vector whose i^{th} coordinate is the prob. that the surfer is at page i at time t
- So, $\mathbf{p}(t)$ is a probability distribution over pages

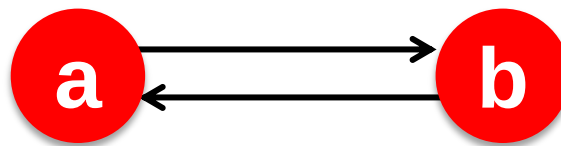
PageRank: The Google Formulation

PageRank: Three Questions

$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i} \quad \text{or equivalently} \quad r = Mr$$

- Does this converge?
- Does it converge to what we want?
- Are results reasonable?

Does this converge?



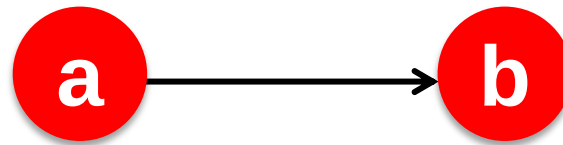
$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i}$$

■ Example:

$$\begin{array}{l} r_a \\ r_b \end{array} = \begin{array}{cccc} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array}$$

Iteration 0, 1, 2, ...

Does it converge to what we want?



$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i}$$

■ Example:

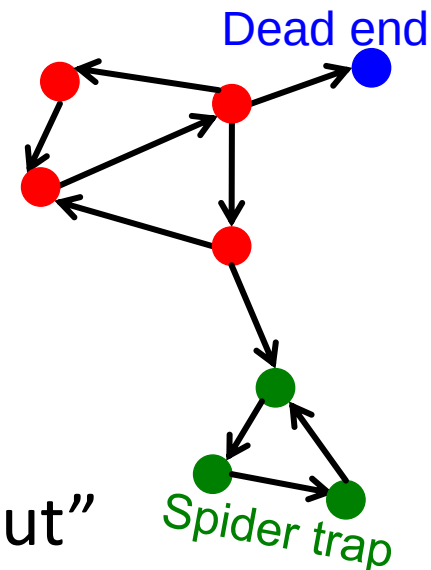
$$\begin{matrix} r_a \\ r_b \end{matrix} = \begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$$

Iteration 0, 1, 2, ...

PageRank: Problems

2 problems:

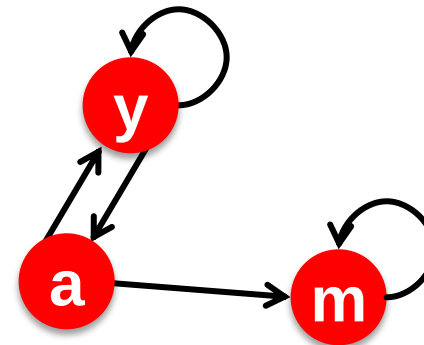
- **(1) Some pages are dead ends** (have no out-links)
 - Random walk has “nowhere” to go to
 - Such pages cause importance to “leak out”
- **(2) Spider traps:** (all out-links are within the group)
 - Random walked gets “stuck” in a trap
 - And eventually spider traps absorb all importance



Problem: Spider Traps

■ Power Iteration:

- Set $r_j = 1$
- $r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
 - And iterate



m is a spider trap

	y	a	m
y	1/2	1/2	0
a	1/2	0	0
m	0	1/2	1

$$\mathbf{r}_y = \mathbf{r}_y/2 + \mathbf{r}_a/2$$

$$\mathbf{r}_a = \mathbf{r}_y/2$$

$$\mathbf{r}_m = \mathbf{r}_a/2 + \mathbf{r}_m$$

■ Example:

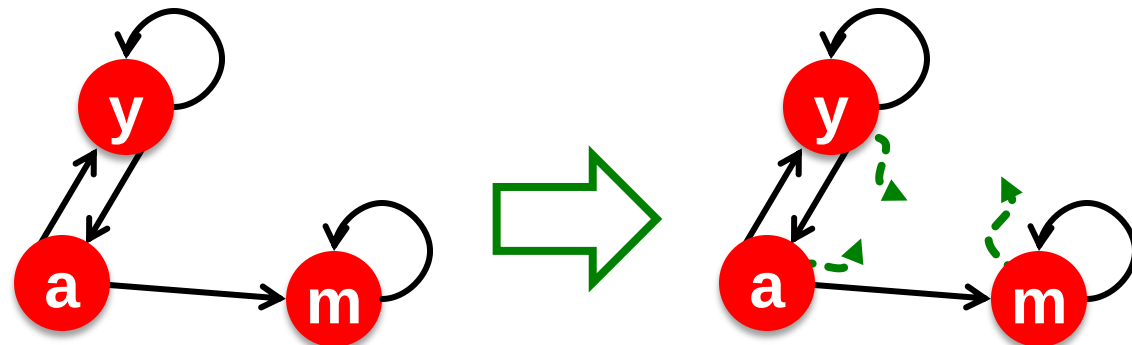
$$\begin{bmatrix} r_y \\ r_a \\ r_m \end{bmatrix} = \begin{bmatrix} 1/3 & 2/6 & 3/12 & 5/24 & & 0 \\ 1/3 & 1/6 & 2/12 & 3/24 & \dots & 0 \\ 1/3 & 3/6 & 7/12 & 16/24 & & 1 \end{bmatrix}$$

Iteration 0, 1, 2, ...

All the PageRank score gets “trapped” in node m.

Solution: Teleports!

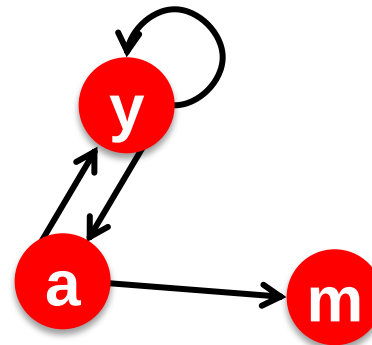
- The Google solution for spider traps: **At each time step, the random surfer has two options**
 - With prob. β , follow a link at random
 - With prob. $1-\beta$, jump to some random page
 - Common values for β are in the range 0.8 to 0.9
- **Surfer will teleport out of spider trap within a few time steps**



Problem: Dead Ends

■ Power Iteration:

- Set $r_j = 1$
- $r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
 - And iterate



	y	a	m
y	1/2	1/2	0
a	1/2	0	0
m	0	1/2	0

$$\mathbf{r}_y = \mathbf{r}_y/2 + \mathbf{r}_a/2$$

$$\mathbf{r}_a = \mathbf{r}_y/2$$

$$\mathbf{r}_m = \mathbf{r}_a/2$$

■ Example:

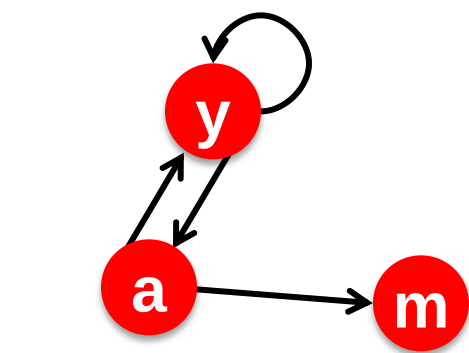
$$\begin{bmatrix} \mathbf{r}_y \\ \mathbf{r}_a \\ \mathbf{r}_m \end{bmatrix} = \begin{array}{cccccc} 1/3 & 2/6 & 3/12 & 5/24 & & 0 \\ 1/3 & 1/6 & 2/12 & 3/24 & \dots & 0 \\ 1/3 & 1/6 & 1/12 & 2/24 & & 0 \end{array}$$

Iteration 0, 1, 2, ...

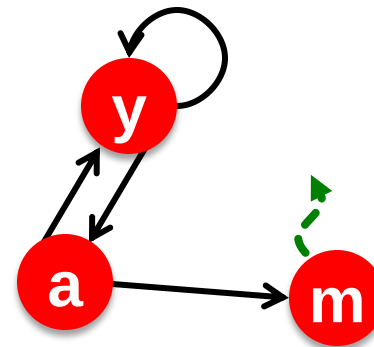
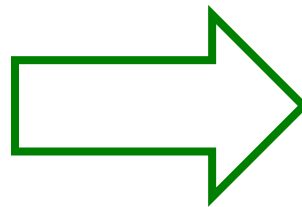
Here the PageRank “leaks” out since the matrix is not stochastic.

Solution: Always Teleport!

- **Teleports:** Follow random teleport links with probability 1.0 from dead-ends
 - Adjust matrix accordingly



	y	a	m
y	$\frac{1}{2}$	$\frac{1}{2}$	0
a	$\frac{1}{2}$	0	0
m	0	$\frac{1}{2}$	0



	y	a	m
y	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{3}$
a	$\frac{1}{2}$	0	$\frac{1}{3}$
m	0	$\frac{1}{2}$	$\frac{1}{3}$

Why Teleports Solve the Problem?

Why are dead-ends and spider traps a problem and **why do teleports solve the problem?**

- **Spider-traps** are not a problem, but with traps PageRank scores are **not** what we want
 - **Solution:** Never get stuck in a spider trap by teleporting out of it in a finite number of steps
- **Dead-ends** are a problem
 - The matrix is not column stochastic so our initial assumptions are not met
 - **Solution:** Make matrix column stochastic by always teleporting when there is nowhere else to go

Solution: Random Teleports

- Google's solution that does it all:

At each step, random surfer has two options:

- With probability β , follow a link at random
- With probability $1-\beta$, jump to some random page

- **PageRank equation** [Brin-Page, 98]

$$r_j = \sum_{i \rightarrow j} \beta \frac{r_i}{d_i} + (1 - \beta) \frac{1}{N}$$

d_i ... out-degree of node i

This formulation assumes that M has no dead ends. We can either preprocess matrix M to remove all dead ends or explicitly follow random teleport links with probability 1.0 from dead-ends.

The Google Matrix

- **PageRank equation** [Brin-Page, '98]

$$r_j = \sum_{i \rightarrow j} \beta \frac{r_i}{d_i} + (1 - \beta) \frac{1}{N}$$

- **The Google Matrix A :**

$[1/N]_{N \times N}$... N by N matrix
where all entries are $1/N$

$$A = \beta M + (1 - \beta) \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{N \times N}$$

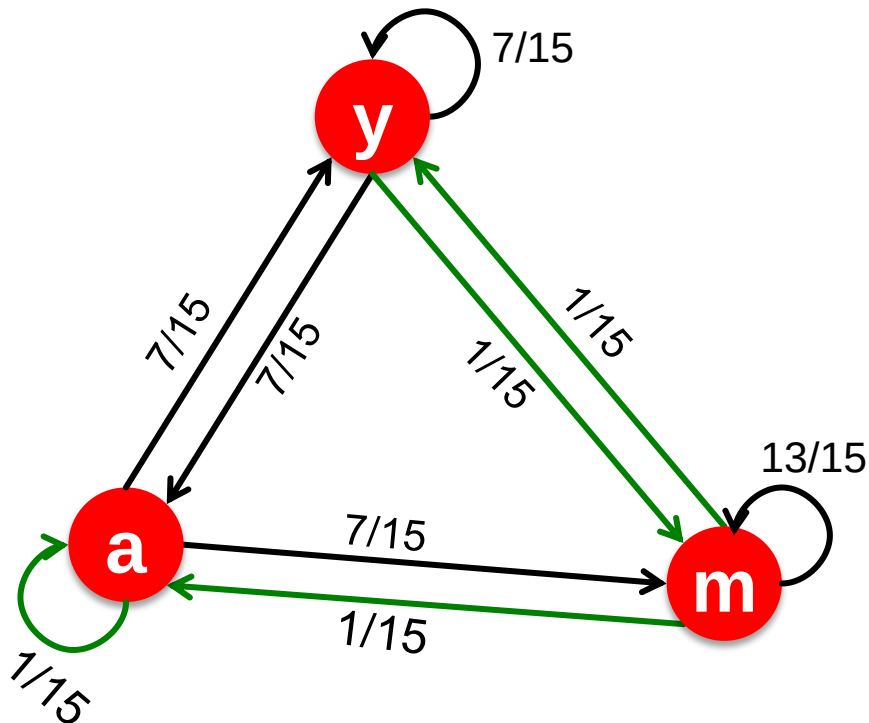
- **We have a recursive problem: $\mathbf{r} = A \cdot \mathbf{r}$**

And the Power method still works!

- **What is β ?**

- In practice $\beta = 0.8, 0.9$ (make 5 steps on avg., jump)

Random Teleports ($\beta = 0.8$)



$$0.8 \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 \\ 0 & 1/2 & 1 \end{bmatrix} + 0.2 \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$$

$$\begin{matrix} y \\ a \\ m \end{matrix} \begin{bmatrix} 7/15 & 7/15 & 1/15 \\ 7/15 & 1/15 & 1/15 \\ 1/15 & 7/15 & 13/15 \end{bmatrix}$$

A

$$\begin{matrix} y \\ a \\ m \end{matrix} = \begin{matrix} 1/3 & 0.33 & 0.24 & 0.26 & & 7/33 \\ 1/3 & 0.20 & 0.20 & 0.18 & \dots & 5/33 \\ 1/3 & 0.46 & 0.52 & 0.56 & & 21/33 \end{matrix}$$

Some Problems with PageRank

- **Measures generic popularity of a page**
 - Will ignore/miss topic-specific authorities
 - **Solution:** Topic-Specific PageRank (**next**)
- **Uses a single measure of importance**
 - Other models of importance
 - **Solution:** Hubs-and-Authorities
- **Susceptible to Link spam**
 - Artificial link topographies created in order to boost page rank
 - **Solution:** TrustRank